

Department of Control Science and Engineering
University of Science and Technology Beijing
No. 30 Xueyuan Road, Haidian District
Beijing 100083, China

August 17, 2026

Editor-in-Chief
Computers & Industrial Engineering

Dear Editor-in-Chief,

Please consider our Research Paper entitled “Two-stage MILP scheduling and structure-aware learning-enhanced branch-and-cut for dual-source mixed-flow rotating-chamber cluster tools” for publication in *Computers & Industrial Engineering*.

The paper addresses an industrial scheduling problem arising in rotating-chamber cluster tools used in a PECVD process setting. Product wafers and reusable vacuum-zone dummy wafers share chambers, load locks, and robots while obeying recipe compatibility, residency, cleaning, capacity, and transfer constraints. Their concurrent interaction creates a finite-horizon scheduling problem that is not captured by single-flow cluster-tool formulations.

The paper makes two connected contributions. First, it develops a finite-horizon two-stage MILP that integrates canonical same-recipe pairing and slot incidence with transfers, cleaning epochs, reusable dummy-wafer allocation, shared-resource ordering, residence limits, and event timing. The canonical incidence removes equivalent labels; the second stage preserves the stage-one discrete schedule and makespan within numerical tolerance while refining physical route waits and processing cadence. Second, it develops RDMCT-HBS, a structure-aware learning-enhanced branch-and-cut method combining a precomputed SPBS incumbent, reconstructed variable-role features, A3C-inspired parallel policy-gradient training, bounded beam decoding, and role-submodular completion.

The computational evidence is both extensive and mechanism-specific. In a 2700-run nominal comparison, RDMCT-HBS records the lowest observed mean primal-dual integral and mean solver time among seven methods. A separate 960-run paired experiment shows that supplying a precomputed SPBS incumbent reduces mean primal-dual integral by 26.19% and increases incumbent availability from 51.67% to 100% during the timed SCIP stage; offline SPBS construction is excluded from that budget. In 1500 matched runs under the same 600-s total budget, the complete two-stage workflow returns schedules with physical route-total wait, route-maximum wait, cadence coefficient of variation, and cadence deviation lower by 51.91%, 68.89%, 39.70%, and 94.05%, respectively, than the no-refinement workflow. A complete 300-run factorial experiment further shows that wafer count and mixed-flow recipe composition dominate the observed difficulty variation. The larger-scale stress test adds 729 time-limited runs on nine 40–64-wafer configurations, every one of which returns an independently validated incumbent. Across the complete evidence programme, all 6273 planned runs are accounted for and all 6041 saved incumbents pass independent feasibility checking against their frozen generated MILPs.

The work fits the journal because it integrates a new industrial-engineering model with an explainable, exact-optimization-compatible learning method and evaluates both computational and manufacturing consequences. The formulation and solution architecture are also transferable to other tightly coupled multi-resource scheduling systems with auxiliary flows and shared transport.

Thank you for your consideration.

Sincerely,

Lizhen Shao
Corresponding author
University of Science and Technology
Beijing
lshao@ustb.edu.cn